

MORNING

27 DEC 2011

Please check that this question paper contains 9 questions and 2 printed pages within first ten minutes.

[Total No. of Questions: 09]

[Total No. of Pages: 02]

Uni. Roll No.

Program/ Course: B.Tech (Batch 2018 onward)

Semester: 3rd

Name of Subject: Mathematics-III

Subject Code: BSCS-101

Paper ID: 16013

Scientific Calculator is Allowed

Time Allowed: 3 Hours

Max. Marks: 60

NOTE:

- 1) Parts A and B are compulsory.
- 2) Part -C has Two questions Q8 and Q9. Both are compulsory, but with internal choice.
- 3) Any missing data may be assumed appropriately.

Part -A

[Marks: 02 each]

Q1.

- a) Separate $\sinh z$ into real and imaginary parts.
- b) Solve the system of equations : $2x - y = -5$, $x - 3y = -30$.
- c) Evaluate $\int_C \frac{dz}{(z+2)^2}$ where C is the circle $|z+i| = \sqrt{2}$.
- d) Write algorithm for straight line fit of a curve.
- e) Check the correctness of the statement, "Mean of a binomial distribution is 3 and variance is 3/2."
- f) In a hospital, 475 female and 525 male babies were born in a week. Do these figures confirm the hypothesis that male and female births are born in equal number.
(tabulated value = 1.96)

Part -B

[Marks: 04 each]

Q2. Fit the curve $y = ax + b$ to the following data :

x	2	5	6	9	11
y	2	4	6	9	10

Q3. Expand $f(z) = \frac{z}{(z+2)(z+3)}$ in the region $2 < |z| < 3$.

Q4. Solve the following equations by Gauss Elimination method :

$$2x + 4y - 6z = -4, \quad x + 5y + 3z = 10, \quad x + 3y + 2z = 5.$$

Q5. An ambulance service company claims that on average it takes 20 minutes between a call for ambulance and the patient arrival at the hospital. If in 6 calls the time taken (between a call and arrival at hospital) are 27, 18, 26, 15, 20, 32. Can the company claim be accepted?
(tabulated value = 2.571)

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Q6. The probability that a pen manufactured by a company will be defective is $1/10$. If 12 such pens are manufactured, find the probability that at least two pens will be defective.

Q7. If $f(z) = u + iv$ is an analytic function and $u = e^x(x \cos y - y \sin y)$, find $f(z)$.

Part -C

[Marks: 12 each]

Q8.(a) Evaluate $\int_0^{2\pi} \frac{\cos 2\theta}{(5 + 4 \cos \theta)} d\theta$ using contour integration.

OR

(b) Evaluate $\int_C \frac{3z^2 + z + 1}{(z^2 - 1)(z + 3)} dz$, where C is $|z - i| = 2$, using Cauchy residue theorem.

Q9.(a) Fit a Poisson distribution to the following data and test the goodness of fit :

x	0	1	2	3	4
f	211	90	19	5	0

(tabulated value = 7.82)

OR

(b) i) A box contains 16 bulbs out of which 4 bulbs are defective. 3 bulbs are drawn one by one without replacement. Find the mean of the probability distribution of the number of defective bulbs drawn.

ii) The mean yield of two sets of plots and their variability are as given. Examine whether the difference in the variability in yields is significant?

	Set of 40 plots	Set of 60 plots
Mean yield per plot	1258	1243
S.D. per plot	34	28

(tabulated value = 1.96)
